TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Image-Based AI Decision Support for Biomedical and Life Science Applications**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (X), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Intelligent Interpretation Systems	Medical Bio	Other Biopharmaceuticals / Bio-Materials
Concept of Proposed Technology	The proposed technology focuses on the development of **AI-based decision support systems derived from biomedical imaging data**, leveraging modern deep learning techniques to extract biologically meaningful information directly from images. In many biomedical and life science applications, large volumes of microscopy or medical images are generated, yet their analysis still relies heavily on manual inspection, handcrafted image features, or narrowly specialized algorithms that do not generalize well across experiments, instruments, or sites. Image-based artificial intelligence provides a scalable solution to this challenge by learning feature representations directly from pixel-level data. Instead of relying on predefined morphological descriptors, the proposed technology learns **compact and expressive feature embeddings from biomedical images** that capture complex cellular or tissue phenotypes relevant for downstream analysis and decision support. The proposed system is explicitly designed as a **decision-support tool**, assisting researchers, clinicians, or industrial users in interpreting image-derived patterns and trends, rather than acting as a fully autonomous diagnostic system. This positioning supports expert oversight, regulatory acceptance, and flexible integration into existing workflows. Intended application areas include, but are not limited to: • Biomedical and pharmaceutical research • Phenotyping and screening of microscopy images			

	• Exploratory diagnostics and stratification support • Functional assessment of biological states under different conditions
Rationale for Joint R&D Project	The proposed joint R&D project aims to combine advanced image-based artificial intelligence methodologies with real-world biomedical and clinical application contexts to develop robust, interpretable decision-support systems derived from biomedical imaging data. While recent advances in deep learning - particularly transformer-based vision models - enable the extraction of rich feature representations from images, their successful translation into practice requires close collaboration between AI technology developers and domain-specific industrial partners. In particular, successful deployment depends on the careful adjustment and preprocessing of raw imaging data to ensure stable, efficient, and reproducible model training. This includes systematically addressing challenges such as variability in imaging protocols, heterogeneity in data quality, and the presence of artefacts or noise, all of which can substantially affect the robustness, generalization, and reliability of AI-based decision-support models. A joint R&D framework is essential to ensure that the developed AI methods address **practical** constraints of biomedical imaging**,** including heterogeneous data sources, imaging variability, limited annotations, and the need for interpretability and robustness. Korean industry partners contribute critical assets such as access to diverse biomedical imaging datasets, clearly defined application scenarios, and operational requirements for deployment in research or healthcare environments. In contrast, the research institute provides expertise in computer vision architectures, representation learning, data augmentation, and batch-effect mitigation**,** ensuring scientific rigor and methodological innovation. Furthermore, collaborative development enables early consideration of regulatory, usability, and integration aspects**,** positioning the resulting technology as a human-in-the-loop decision-support system rather than an autonomous diagnostic solution. This alignment reduces translational risk and supports adoption in regulated biomedical and digital health contexts. The project also aligns strongly with Korea's national strategy to strengthen **AI-driven healthcare and life science innovation**, where multiple SMEs and research hospitals are actively investing in medical imaging AI and decision-support technologies. Through joint R&D, the project accelerates technology transfer, enhances the competitiveness of Korean SMEs, and supports the creation of scalable, regulation-ready AI solutions for biomedical research and healthcare applications. Potential Korean partners include AI and healthcare companies with strong imaging expertise, such as Lunit**,** VUNO**,** Deep Bio**,** Mediwhale**,** Tomocube, and INFINITT Healthcare.
Key Words	1. Image-based decision support 2. Biomedical image phenotyping 3. Vision Transformers (ViT)

	4. Representation learning 5. Robust and interpretable AI

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3-4	End of the Project	Level 6
R&D Objectives	The final goal of the project is to deliver a robust, scalable, and interpretable AI-based image decision support framework that transforms biomedical images into biologically meaningful feature embeddings and actionable decision-support outputs, suitable for integration into industrial and research workflows. Main Objectives • Development of a standardized pipeline for biomedical image ingestion and preprocessing • Learning of compact feature embeddings from images using transformer-based vision models • Integration of data augmentation and batch-effect mitigation strategies to ensure robustness • Design of decision-support models that aggregate image-derived features into higher-level outputs • Validation of model performance, stability, and interpretability across heterogeneous datasets			
R&D Target Specifications	**1. Image Representation and Feature Embedding** • Development of computer vision models capable of learning compact feature embeddings directly from biomedical images, avoiding reliance on handcrafted descriptors that require manual parameter tuning and are often insufficiently tailored to the specific data distribution, imaging setup, or biological context. • Feature embeddings shall preserve **biologically meaningful morphological and spatial information**, enabling downstream phenotypic analysis and comparison across samples and conditions. • Target embedding representations are designed to be **compact, reusable, and modality-agnostic**, supporting integration into multiple analytical pipelines. **2. Robustness and Generalization** • Implementation of **systematic data augmentation and batch-effect mitigation strategies** to ensure stable performance across imaging batches, instruments, and experimental conditions. • Demonstrated robustness against non-biological variability, with consistent embedding distributions across independent datasets.			

	• Validation across heterogeneous datasets to verify generalization beyond the training domain. **3. Decision Support Performance** • Delivery of **image-derived decision-support outputs** (e.g., scores, rankings, phenotype summaries) suitable for research and exploratory clinical contexts. • Performance targets include reproducible differentiation between biological conditions or experimental groups, assessed using established statistical and machine learning metrics. • The system is explicitly designed to support expert interpretation rather than fully autonomous decision-making. **4. Interpretability and Transparency** • Integration of interpretability mechanisms (e.g., attention-based aggregation and embedding-level analysis) to provide transparent insights into which image-derived features contribute to decision-support outputs, including mechanisms to localize and visualize the image regions or segments that most strongly influence model predictions. • Output shall be presented in a form accessible to domain experts, enabling biological or clinical plausibility assessment. **5. Software Quality and Integration Readiness** • Delivery of **well-documented, reproducible, and modular software components** suitable for integration into industrial R&D or digital health workflows. • Support for scalable deployment in research and pilot environments, with clear interfaces for data ingestion, model inference, and result export.
R&D Plan	**Image Acquisition and Preprocessing** The system is designed to operate on high-content biomedical imaging data, such as microscopy images acquired across multiple channels, experimental batches, or imaging devices. Preprocessing steps include normalization of intensity distributions, harmonization of imaging channels, and standardized cropping or tiling strategies to ensure consistent input to the learning models. **Feature Embedding from Biomedical Images** At the core of the approach is the extraction of feature embeddings from biomedical images using Vision Transformer (ViT) architectures or closely related transformer-based vision models. These

	models learn rich representations directly from image patches, enabling the capture of subtle morphological and spatial patterns that are difficult to encode using handcrafted features. The learned embeddings are compact, information-rich representations that summarize the relevant biological state of the imaged object (e.g., cell, tissue region), forming the basis for downstream analysis and decision support. **Model Training and Validation Strategy** Model training follows a data-driven representation learning paradigm, emphasizing robustness and generalization. Data augmentation strategies are systematically applied to account for imaging variability, technical batch effects, and experimental noise. These include intensity transformations, spatial augmentations, and augmentation schemes designed to promote invariance to non-biological factors. Validation is performed across independent datasets and experimental conditions to assess generalization performance and stability, ensuring that the learned representations retain biological signal rather than technical artifacts. **Decision Support Outputs and Interpretability** The extracted image embeddings are aggregated and analyzed to generate decision-support outputs such as risk scores, phenotype summaries, or comparative metrics between conditions. Interpretability mechanisms, including attention-based aggregation or embedding-level analyses, are used to provide transparency into which image-derived patterns contribute most strongly to the decision-support output. Importantly, the system is designed to support expert decision-making, allowing users to contextualize AI-derived insights rather than replacing expert judgment.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Design and development of the AI architecture for image-based representation learning • Adaptation of Vision Transformer models to biomedical imaging constraints • Development of robust training and augmentation strategies addressing batch effects • Validation of model robustness, interpretability, and biological relevance • Knowledge transfer to Korean partners through documentation, training, and joint evaluations
Expected R&R of Korean SME Partner	• Provision of real-world biomedical imaging data and application requirements • Definition of practical use cases and decision-support needs • Participation in validation and demonstration of prototype systems • Feedback on usability, integration, and industrial relevance • Support for downstream adoption and potential commercialization pathways

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes** • A validated AI framework for image-based feature embedding using transformer-based vision models • Robust representation learning pipelines resilient to imaging variability and batch effects • Interpretable decision-support outputs grounded in image-derived biological features **Industrial and Commercialization Relevance** • Transferable AI components suitable for integration into biomedical R&D and digital health tools • Reduced dependence on handcrafted image features and manual analysis • Accelerated image-based analysis workflows for industrial and research users **Impact on Korean SMEs and Life Science Innovation** The project will enable Korean SMEs to adopt state-of-the-art image-based AI decision support technologies without the need to develop foundational AI methods internally. By strengthening capabilities in biomedical image analytics, the project supports innovation in life sciences, digital health, and AI-driven research tools, contributing to increased competitiveness and international collaboration opportunities.

6. References

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(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization